p_E density closure note (lead 2, no new runs)

2026-10-27. Branch `claude/frobenius-symmetry-m-sweep-20261027`. Analysis of recorded values only: the input is `research/twist_obstruction_20261026/reports/run-20261026T1456Z/rows.jsonl` (4969 rows, 4750 `A_random` decomposition targets; not moved to an artifact store). The tool is `src/main.rs` (std + serde_json), its outputs are `closure.json`, `groups.tsv`, `analysis_stdout.txt`. Claim class for every statement below: **exact identity** (derived), **measured** (from the recorded rows), or **heuristic** (labelled).

What was recomputed (measured)

For each of 40 groups (n, l, m, curve; n = 11, 13, 17, 19; m = 2, 3; five curves each) the tool recomputes $p_E = v_E/(2^l - 1)$ and the mean solutions per `A_random` target. 142 of 4750 targets are flagged by the filter (x_R in V, or any off-parity solution recorded): 121 at m = 2 and 21 at m = 3. Only the 21 m = 3 targets are contaminated in the sense that matters: they carry the degenerate family (x_R, x, x) of about 2^l solutions whose size does not depend on p_E (per-curve counts at m = 3 reach 71 to 104 at n = 11 and 23 to 180 at n = 13 in the recorded totals). The 121 m = 2 flags have x_R in V but no off-parity rows and solution counts 0 to 9: no degenerate family exists at m = 2, so the filter drops 121 benign targets there (harmless to the means, but they are not outliers). Flagged targets are reported separately; the "clean" mean excludes them. The recorded per-curve means include them (for example n = 11, m = 3, `rand2`: 15.07 recorded, 10.96 clean). Rows where `solutions` differs from pure + pure_with_2tor + off + violation: 0 of 4750.

Spread at fixed V (m = 3, clean means): max/min across the five curves is 2.64 (n = 11, l = 6), 1.95 (n = 13), 2.99 (n = 17), 1.63 (n = 19), with p_E from 0.397 to 0.576. This is the "3x".

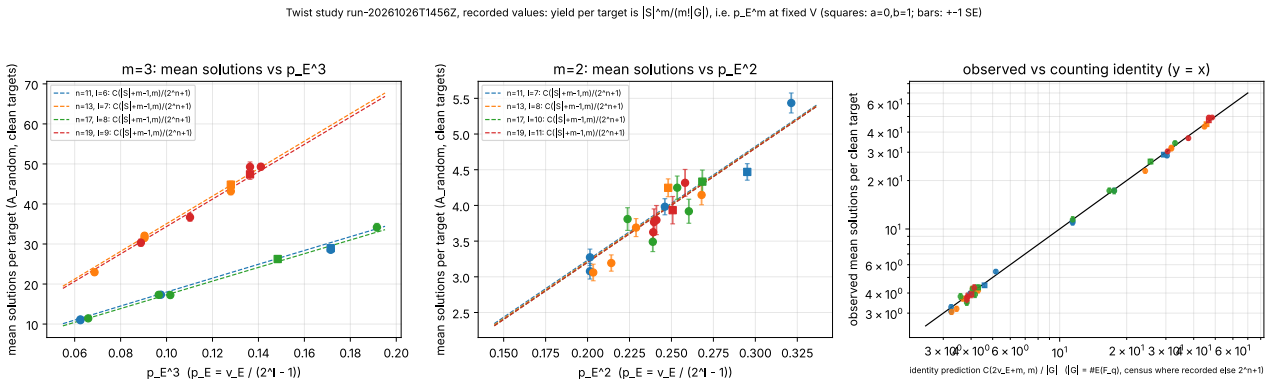


Figure 1: Recorded points (clean targets, bars ± 1 SE) against p_E^m; dashed: the counting identity (section "Counting identity") with $|G| = 2^n + 1$. Right: observed against identity prediction using the census $|G|$ where recorded.

Result (measured):

- Observed / identity: per (n, l, m) cell the mean ratio is 0.97 to 1.02 and its spread across curves is CV 2.0% to 6.6%. Sampling error alone predicts 2.4% to 4.9% (from the recorded SEs). The "constant within about 5%" is therefore at the resolution of the data, not tighter. For n = 11 and 13 the random curves have no recorded $|G|$ (the census used other curves), so $2^n + 1$ is used there, which is off by up to 3% (census range 1998 to 2116 at n = 11).
- Log-log slope of mean on p_E at fixed (n, l), clean means: m = 3: 2.96 $\pm$ 0.06 (15 dof; z against 3: −0.67). m = 2: 2.19 $\pm$ 0.19 (z against 2: +0.99). Recorded totals including degenerate targets: 2.94 $\pm$ 0.13 and 2.17 $\pm$ 0.18.
- Fano factor of the per-target count is 0.4 to 1.3, so counts are not overdispersed on clean targets.

Counting identity (exact)

Let $G = E(F_q)$, $F \subseteq V$ the E-type x-coordinates (|F| = v_E) and $S = \{\pm P_x : x \in F\} \cup \{T\}$, so $|S| = 2v_E + 1$ (x = 0 lifts to the 2-torsion point T). A solution is a multiset of m points of S with sum R. For uniform R in G, orthogonality of characters gives

$$\mathbb{E}_R[N(R)] = \frac{|S|^m}{|G|}, \quad N(R) = |\{(P_1, \ldots, P_m) \in S^m : P_1 + \ldots + P_m = R\}|,$$

since only the trivial character survives the average. Per target, $N(R) = |S|^m/|G| + |G|^{-1} \sum_{\chi \neq 1} \overline{\chi(R)} \hat{S}(\chi)^m$, so the target-to-target variance is controlled by the non-trivial Fourier mass of S and the mean is not. Unordered keys:

divide by m! with the repeat correction, giving the tool's prediction $\binom{|S|+m-1}{m}/|G|$ (the error from sign collisions is of relative order $1/|G|$). With $|S| = 2p_E(2^l - 1) + 1$ and fixed V,

$$\text{yield per target} \propto p_E^m$$

to leading order (the +1 and the repeats give relative corrections of order $1/|S|$, 2 to 4% here). The test is section "What was recomputed": slope 2.96 ± 0.06 at m = 3, ratio to the identity 0.97 to 1.02. Nothing about the curve other than $|G|$ and v_E enters; no isogeny-class, twist or Frobenius structure is visible in the mean, which is the class-invariance already recorded in LEAD-ICE-curve-presentation-invariance ("mean yield is a class invariant") and the set-property statement in LEAD-STR-class-representative-ic-selection (pair46: a pushed crater base gives the crater yield on the floor).

Fixed |F|. Write the identity in $|F|$: yield per target $= \binom{2|F|+m}{m}/\ |G|$ for every curve and every V with that many E-type points. The p_E^m factor disappears: curves with high p_E give more relations at fixed V only because their V holds more points of F, i.e. a larger factor base. At matched |F| and matched yield the curves are equal (measured: panel c of the figure, all five curves per cell on one line). What remains is the cost of reaching |F| points: V must have $2^l = |F|\ /p_E$ elements, so the dimension is $l = \log_2|F| + \log_2(1/p_E)$. Choosing a curve or basis with larger p_E saves at most $\log_2(p_{E'}/p_E)$ bits of dim V, and the absolute ceiling is $p_E = 1$ against the generic 1/2: one bit (a factor 2 in any cost that is $\propto 2^l$, for example the 2^(l+1) eliminations per probe of PinX3). The yield-at-fixed-V gain is $\left(\frac{p'}{p}\right)^m$, at most 8 at m = 3 (same ceiling, $p' = 1$), but that gain is the larger factor base and is paid for at fixed |F| by the dimension. The two ceilings are at different operating points: the 8x is a yield ratio at fixed V, and the relations needed also scale with |F|, so the cost ceiling at fixed V is about $(p'/p)^{m-1}$ (4 in probes at m = 3); the factor 2 assumes cost proportional to 2^l with continuous l, but l is an integer. Both ceilings are adversarial bounds, not achievable values (see "What choosing b..."). The direct matched-|F| evidence is only the few coincident-v_E pairs (n = 11, l = 6: v_E = 25 twice and 35 twice, means 10.96 / 11.16 and 28.95 / 28.55); panel c of the figure is the identity-versus-observed plot, the same test as the ratio check.

What choosing b or an isogenous curve buys at n = 131, l = 44, m = 3

Facts. (a) For $y^2 + xy = x^3 + ax^2 + b$, the parameter b is an F_q-isomorphism invariant (the x-preserving map y -> y + s x changes a by $s^2 + s$; x -> x + s changes a3 and leaves the normal form), so ECC2K-130 (a = 0, b = 1) has no isomorphic model with another b. Only another V, or a different curve in the isogeny class (same $|G|$, same trace), changes p_E. (b) x is E-type iff $\mathrm{Tr}(x + a + b/x^2) = 0$, so $p_E = \frac{1}{2} + S/\big(2(2^l - 1)\big)$ with $S = \sum_{x \in V^*} (-1)^{\mathrm{Tr}(x) + \mathrm{Tr}(a) + \mathrm{Tr}\left(\sqrt{b}/x\right)}$, a Kloosterman-type sum over a subspace.

Heuristic (labelled). If S behaves like a random sum, $\sigma_p = 2^{-l/2-1}$ (1.19e-7 at l = 44). Check on the recorded groups (l = 6 to 11): $z = (2p_E - 1)\sqrt{2^l - 1}$ has pooled RMS 1.23 over 40 groups, per-l RMS 0.79 to 1.54 (not independent: the same five curves per n). The aggregate hides heterogeneity: split by m, the 20 m = 3 groups have RMS 1.45 (sum z^2 = 41.8, p about 0.003) and the m = 2 groups 0.97; (m, l) = (3, 8) has RMS 1.96 with single groups at z = −3.07 and +2.44, far outside the 1-sigma band of Fig. 2. The 40 groups are not independent (V is nested between m = 2 and m = 3 for the same curve). This is consistent with σ_p only to within about 1.5x, which is the edge of the data, and says nothing at l = 44. The "below 10^{-4}" budget claim (5.2e-5 for the all-subspace row) holds only if σ_p is underestimated by less than about 1.9x; a larger underestimate breaks it. No unconditional nontrivial bound is known to me for such sums on a 44-dimensional subspace of $F_{2^{131}}$ (l below n/2), so the extrapolation is a heuristic.

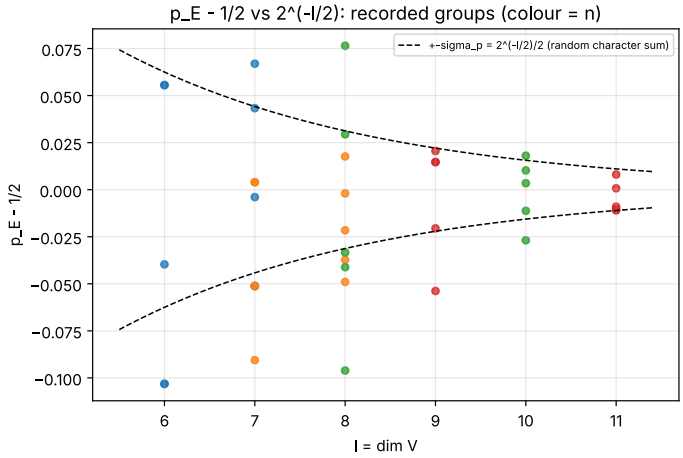


Figure 2: Recorded $p_E - \frac{1}{2}$ against l, with the random-sum band $\pm 2^{-l/2}/2$.

Best case from selection, Gaussian maximum over K independent candidates (`closure.json`, `budget_n131_l44_m3`):

candidates K	z of the best	δp	yield gain at fixed V, m = 3	dim V saved at fixed |F|
2^{10} curves	3.10	3.7e-7	2.2e-6	1e-6 bit
2^{20} curves	4.76	5.7e-7	3.4e-6	1.6e-6 bit
2^{66} (class-size ceiling)	9.23	1.1e-6	6.6e-6	3.2e-6 bit
all 2^{3828} 44-dim subspaces	72.8	8.7e-6	5.2e-5	2.5e-5 bit

A 1% yield gain needs z = 14,000, $\ln K \approx 9.8 \times 10^7$. The class has at most about 2^{66} isomorphism classes, so it cannot supply it. Even the full subspace family (heuristically independent, which it is not) gives 5e-5. Selection also costs more than it returns: evaluating p_E of one candidate means enumerating 2^{44} points of V, and moving the instance to another class member needs a verified isogeny chain, both charged per candidate under the 0, 0.1 and 1 GAE band (§5a) with no effect on the sign of the conclusion: the relative yield gain is below 10^{-4} and the dimension saved is below 10^{-4} bit while the price is at least 2^{44} field operations per candidate. Only a factor base that is not a subspace (F chosen as E-type points, $p_E = 1$ by construction) escapes this, and it is the relabelling of the counting-identity section paid for by losing the linear structure the 2^(l+1)-eliminations method needs. That is a different lead (non-subspace bases), not this one.

Scope and limits

- The data are toy fields (n <= 19, l <= 11, 40 groups, five curves per n), one seed, `A_random` targets drawn uniformly from E(F_q) minus O and T. $|G|$ comes from the census for 4 of the 40 groups (n = 7, 11, 13 only) and is approximated by $2^n + 1$ for the other 36, i.e. 18 distinct curves (n = 11: 4, n = 13: 4, n = 17: 5, n = 19: 5, including the Koblitz curve at n = 17 and 19). The approximation moves predictions by up to about 3% at n = 11 and below 0.6% at n >= 17.
- The identity is exact only for the first moment over R, and for the unconditional mean over all R in G. The clean sample excludes R in S, O and T; to O(1/|G|) its expectation is $\binom{|S|+1}{3}/|G|$ at m = 3, i.e. the stated prediction times $2v_E/(2v_E+3)$ (0.954, 0.977, 0.988, 0.994 for n = 11, 13, 17, 19). The data cannot tell the two apart (cell SEs 1 to 2%), so the "0.97 to 1.02 agreement" is a few-percent check; the slope tests are unaffected. The m = 2 slope SE of 0.19 is conservative (a target-level bootstrap gives about 0.13, z against 2 about 1.5), the slope test has no pre-stated acceptance rule (it is post hoc on data the twist study had already seen), and the expected slope under the identity is about 3 - 6/|S|, indistinguishable from 3 here. It predicts nothing about the cost of the Groebner or elimination step per probe, which the presentation-invariance ICC studies (n = 31 and 23, m = 3) measured separately and found null.
- The large-l statements (section "What choosing b...") are heuristic extrapolations of σ_p; the adversarial ceilings (factor 2 in dimension, 8 in yield) are unconditional but unreachable by any known construction.
- No ECDLP cost was measured. No row for the IC leaderboard or `benchmarks/`; no matched rho is involved, so §5a pricing is stated but not exercised: no priced table exists, and the sentence in section 3 that the 0, 0.1 and 1 GAE band does not change the sign rests only on 2^44 point tests against a gain below 10^{-4}. The n = 131 (m = 83 / 31)

numbers are pure extrapolation from toy fields with n <= 19 (n prime; ord_n(2) = 10, 12, 8, 18 against 130); the closure is not established at the target size, and no ECDLP claim is made.
- Unchecked: whether the 5.8% floor/crater yield deficit and 7.2% model effect recorded in LEAD-STR (prime fields, l = 2 volcanoes) are $(p_{\text{floor}}/p_{\text{crater}})^m$ at unmatched |F|. The identity predicts they are if |F| differs; the recorded |F| per curve was not examined here.

Atlas disposition (not applied; closure pending review)

This note adds evidence. It does not close anything. A closure claim goes to the breakthrough siblings (validator-breakthrough, red-team-breakthrough) in their own sessions (AGENTS.md "Model tiers"); this note is not that review.

- **LEAD-STR-class-representative-ic-selection**: suggested wording is "evidence added; closure of the sub-claim 'pick the isomorphic or isogenous model with the best small-x factor base' pending breakthrough review and the |F| check". The scope that this note supports is narrow: binary subspace bases, first-moment yield, under the random-character-sum heuristic for n = 131 (validated only at l <= 11). It does not cover S (only the first moment), the prime-field volcano evidence of the lead (this note's data are binary toy fields, n <= 19), or the transport and floor/crater arms. The 5.8% floor/crater and 7.2% model effects have not been checked for being |F| relabelling; that free check (compare the recorded |F| of the floor and crater arms; if the yield ratio is explained by $(|F| \text{ ratio})^m$, match the preregistered three-arm study on |F|) should be run first and decides the disposition. The unconditional ceiling that remains is a factor 2 in cost.
- **LEAD-ICE-curve-presentation-invariance**: add this note as evidence, with a correction to the lead's wording: "mean yield is a class invariant" holds only at matched |F| or averaged over V. At fixed V the mean varies up to 2.64 to 2.99x across curves of equal order (Spread paragraph above; the twist study's own REPORT, README and RUN_REPORT do not record this 3x, so this note is its source). The recorded yield ICC nulls (0.006, 0.015) are compatible with the identity (expected curve-ICC about 1e-3 at n = 31, l = 10, estimated by the reviewer, not measured here). Disposition unchanged (`open_next_step`, EXP-DREG-dd668d): Groebner-degree transport is a different mechanism.
- Suggested best_measured entry: "p_E spread at n = 131, l = 44 under the random-sum heuristic: gain below 1e-4 in yield; ceiling one bit of dim V; not established at the target size".